Benchmark Report — California 2021 StationMax Uniform Kernel

Date: 2026-03-13
Dataset: California 2021
Risk map: Pyrologix ignition probability (static, trained on 2006–2020, no data leakage for 2021)

Overview

This benchmark tracks the new **StationMax uniform-kernel** placement strategy. It keeps the same high-level formulation as the greedy-kernel StationMax model:

- coverage from **different charging stations does not add** at a cell
- each cell only receives credit from its **best selected station**
- candidate station-cell pairs are restricted to the masked **one-way reachable** zone with `floor(max_battery_time / 2) = 3` operational cells

The difference is the **same-station multi-drone kernel**:

- for a station with `k` drones and `R` feasible masked reachable cells
- each reachable cell gets coverage `min(1, k * B / R)`
- where `B = max_battery_time = 7` operational substeps

So when all `7 × 7 = 49` cells are feasible, this reduces to `min(1, k / 7)`.

For budgets above `300M`, the objective also includes the small regularization term `-0.1 × budget_used`, which prefers cheaper solutions once coverage is already saturated.

Common Setup

Parameter	Value
Data grid	`1309 × 805` (~1 km/cell)
Operational grid	`261 × 161` (5 km/cell)
Drone speed	`600 m/min`
Coverage radius	`2900 m`
Battery	`1 h = 7` operational substeps
One-way reach used in pruning	`3` operational cells
Candidate pre-filtering	top 20% by risk / coverage potential
Solver	Gurobi
Time limit per run	`600 s` default; `1800 s` for the final 100M rerun

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Fire Locations

500M Run

The `500M` case was executed after introducing the new StationMax uniform strategy.

Observed result:

Metric	Value
Termination	`TIME_LIMIT`
Preprocessing	`2.37 s`
Model creation	`6.30 s`
Solve time	`604.79 s`
Ground sensors	0
Charging stations	0
Drones	0
Budget used	`0.00M / 500.0M`

The solver returned objective `-0.0` with a positive bound, so this should be interpreted as **failure to find a meaningful feasible incumbent within the 10-minute cap**, not as a valid empty optimum.

Plots

Reproduction

To rerun the full benchmark set:

```
bash report/benchmark_2021_stationmax_uniform_kernel/reproduce_benchmark_2021_stationmax_uni
```

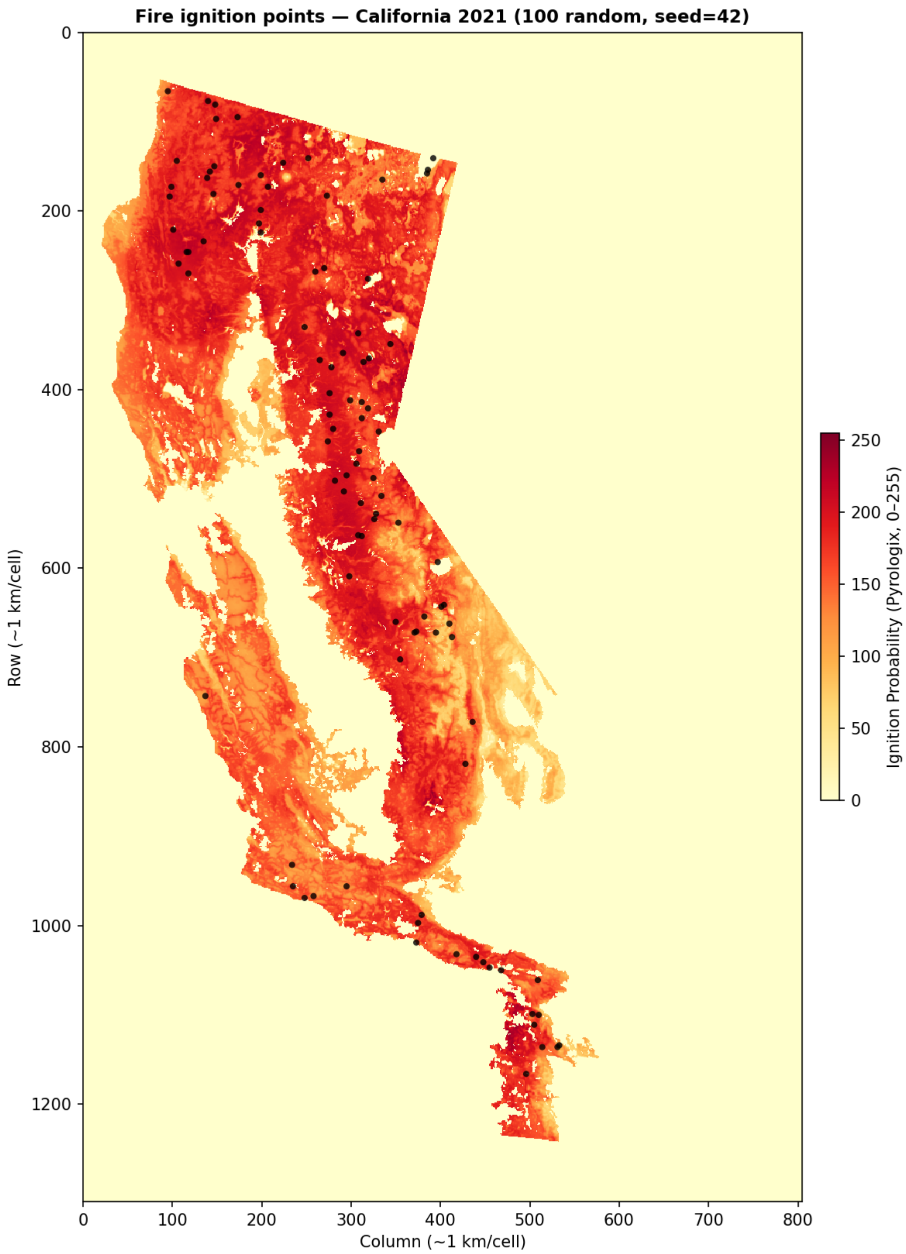

Figure 1: California 2021 fire locations

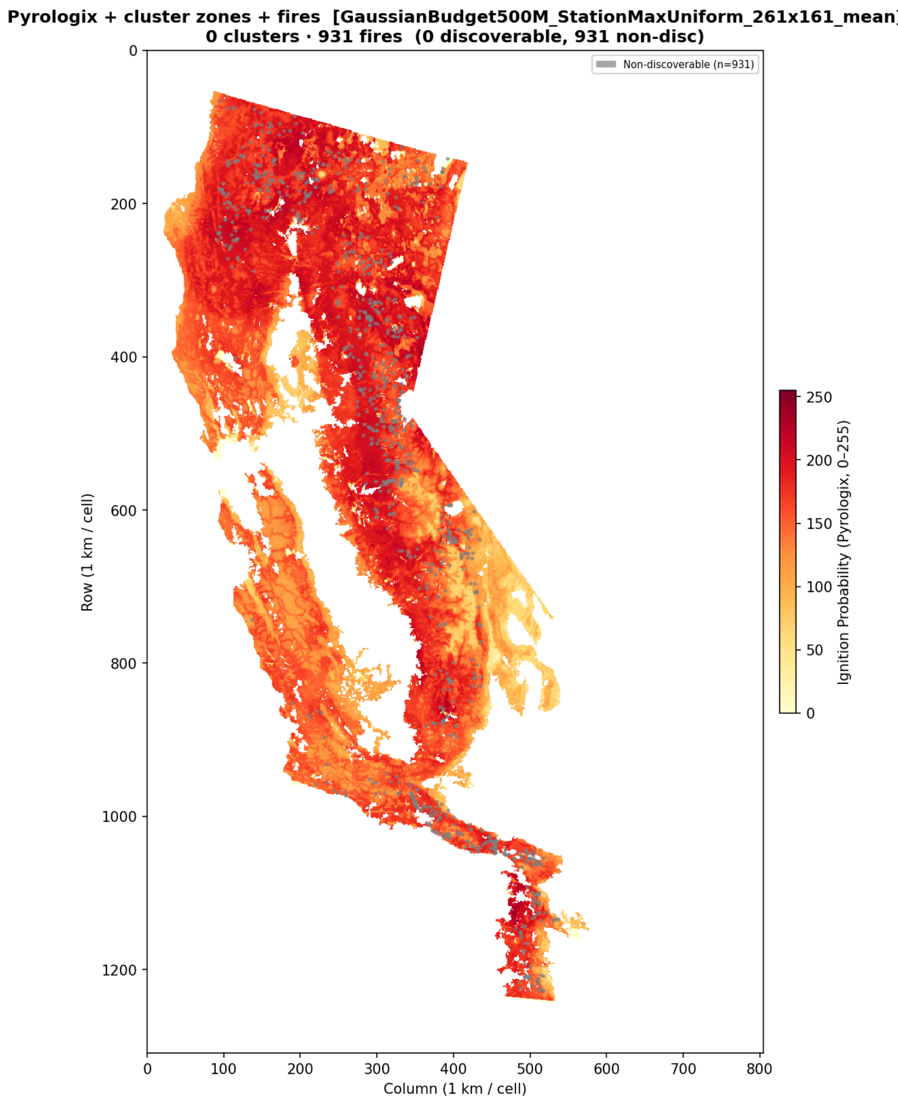

Figure 2: 500M StationMax uniform kernel — data scale

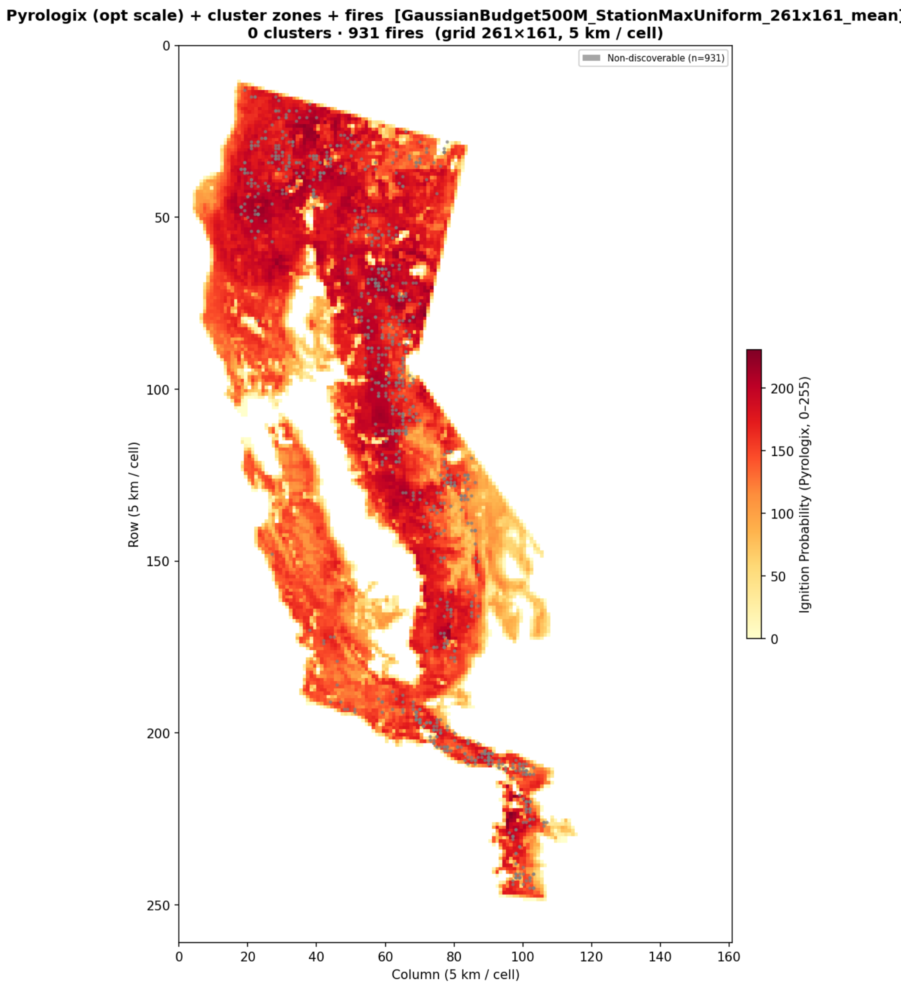

Figure 3: 500M StationMax uniform kernel — operational scale